ROAD DAMAGE DETECTION AND CLASSIFICATION



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INTRODUCTION & PROBLEM STATEMENT

PROBLEM

- Road damage creates hazards and increases maintenance costs.
- Traditional manual inspection is expensive, slow, and inconsistent

SOLUTION

- Automated deep learning object detection.
- Simultaneously localizes (bounding boxes) and classifies damage types.

CONTRIBUTIONS

- 3-architecture comparison (SSD → Faster R-CNN → YOLOv8s).
- Input-resolution ablation study.
- Deployed Gradio web application for real-time inference

METHODOLOGY

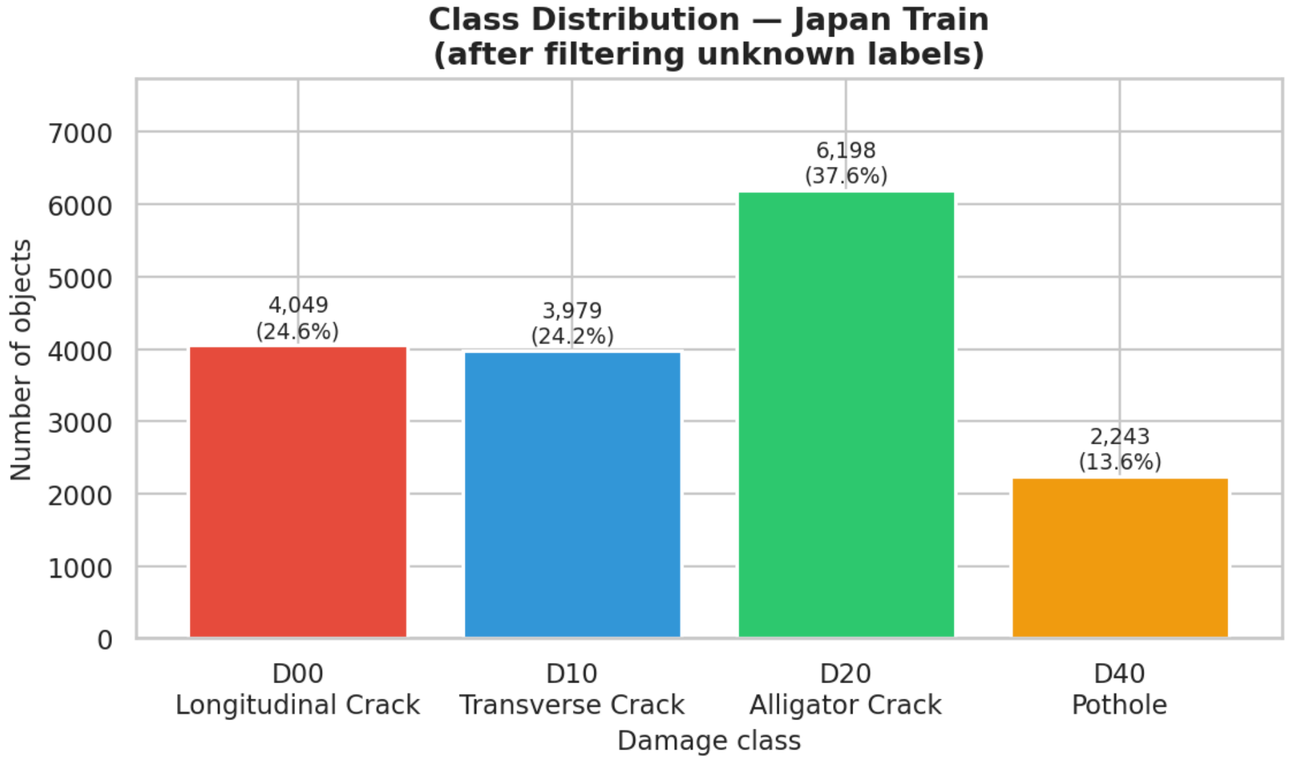
DATASET

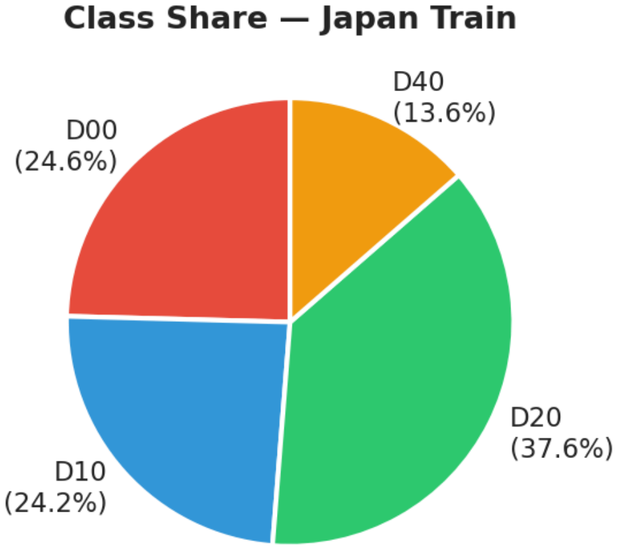
- Used the RDD2022 dataset, specifically the Japan subset
- Focused on 4 damage classes: D00, D10, D20, and D40
- Removed irrelevant/invalid labels before training
- Kept background-only images to reduce false positives

MODEL DEVELOPMENT

- Followed an iterative, evidence-driven approach to improve detection step by step.
- Developed the models in sequence: SSD → Faster R-CNN → YOLOv8s, with an additional input-resolution ablation study.

METHODOLOGY





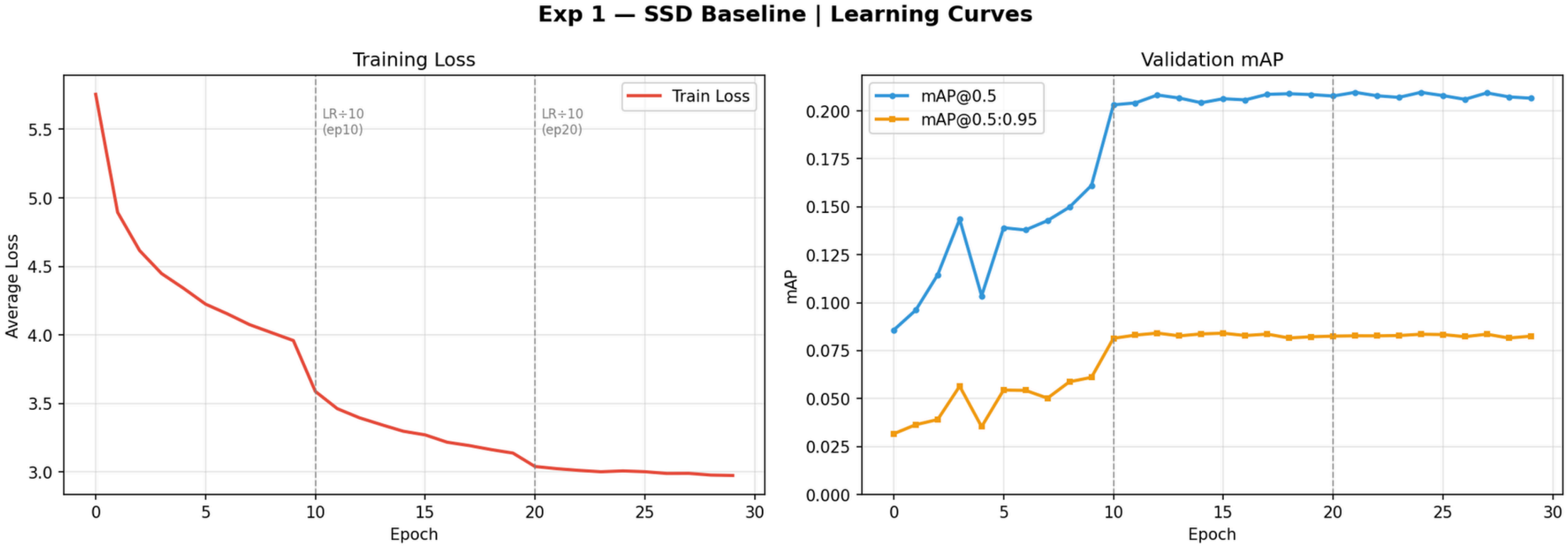
Total valid objects: 16,469 | D40 (Pothole) is underrepresented at 13.6%

EXPERIMENTS

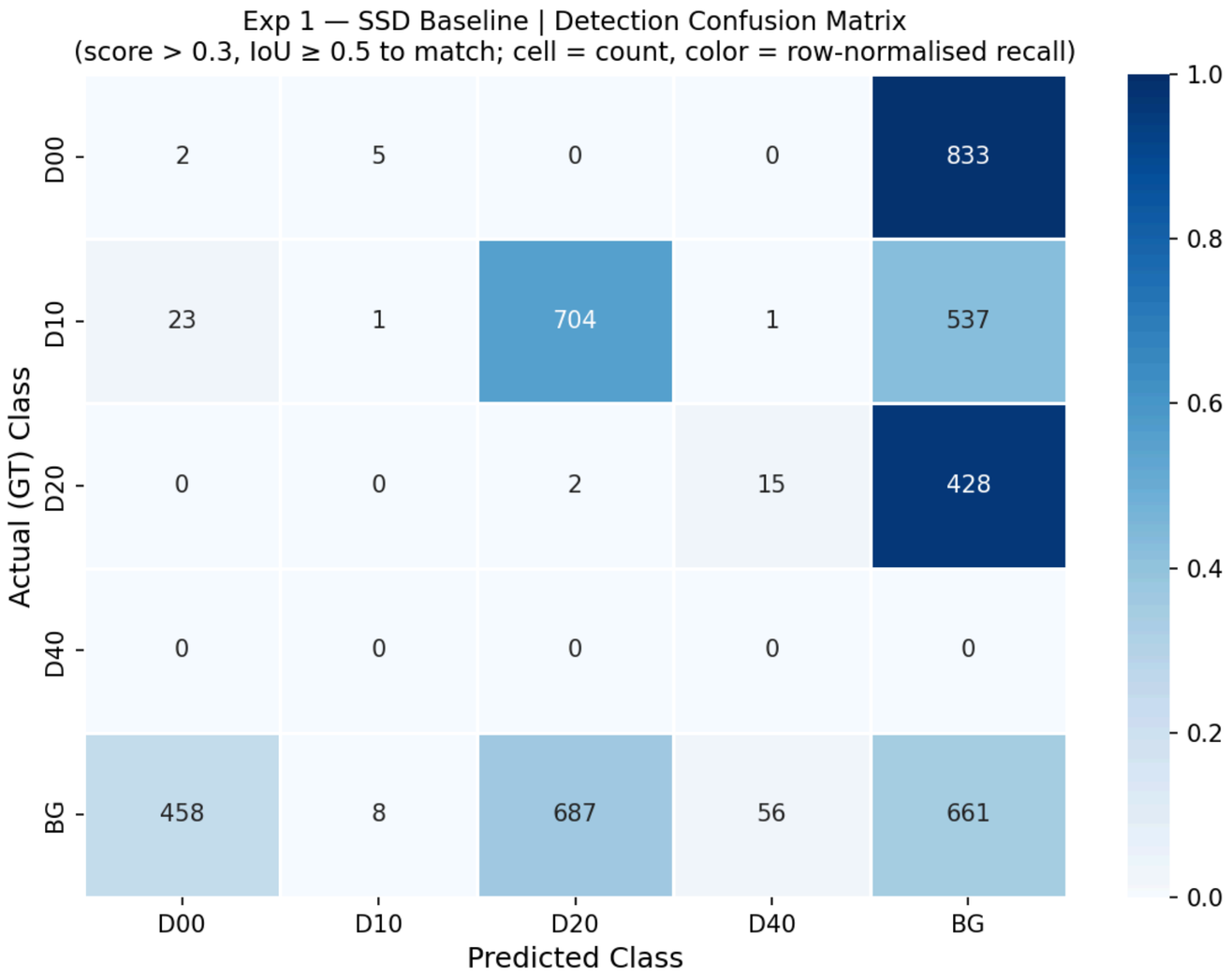
EXPERIMENT 1: SSD BASELINE.

- Model: SSDLite320 + MobileNetV3-Large
- Purpose: establish a weak but fast baseline
- Best result: mAP@0.5 = 0.2097
- Main issues: underfitting, imprecise localization, insufficient resolution

EXPERIMENT 1



EXPERIMENT 1



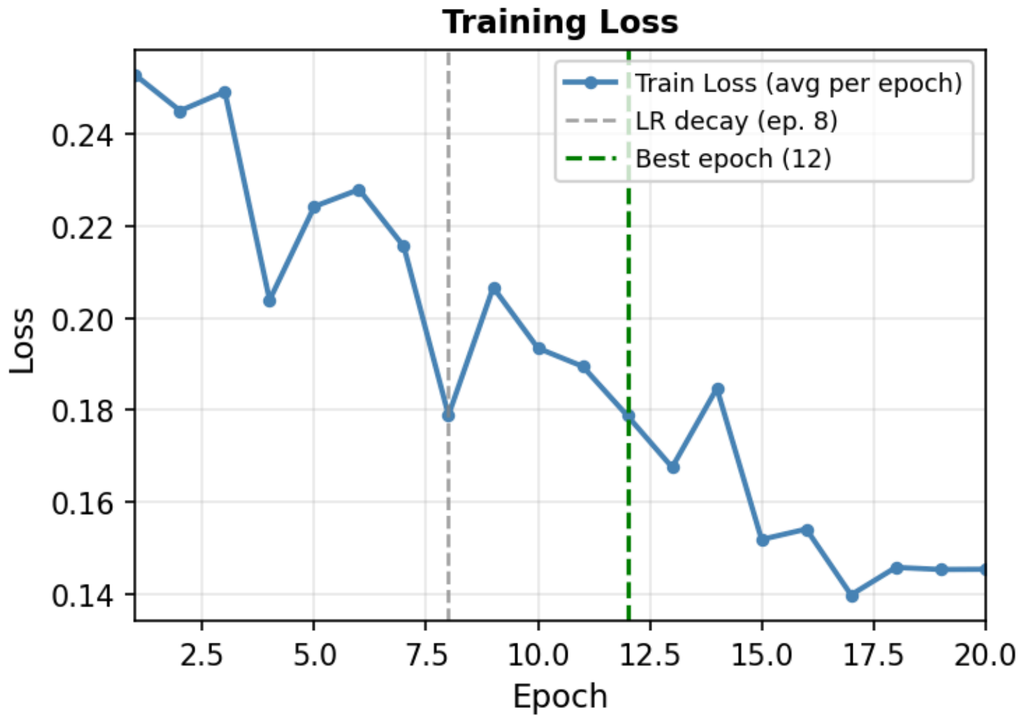
EXPERIMENTS

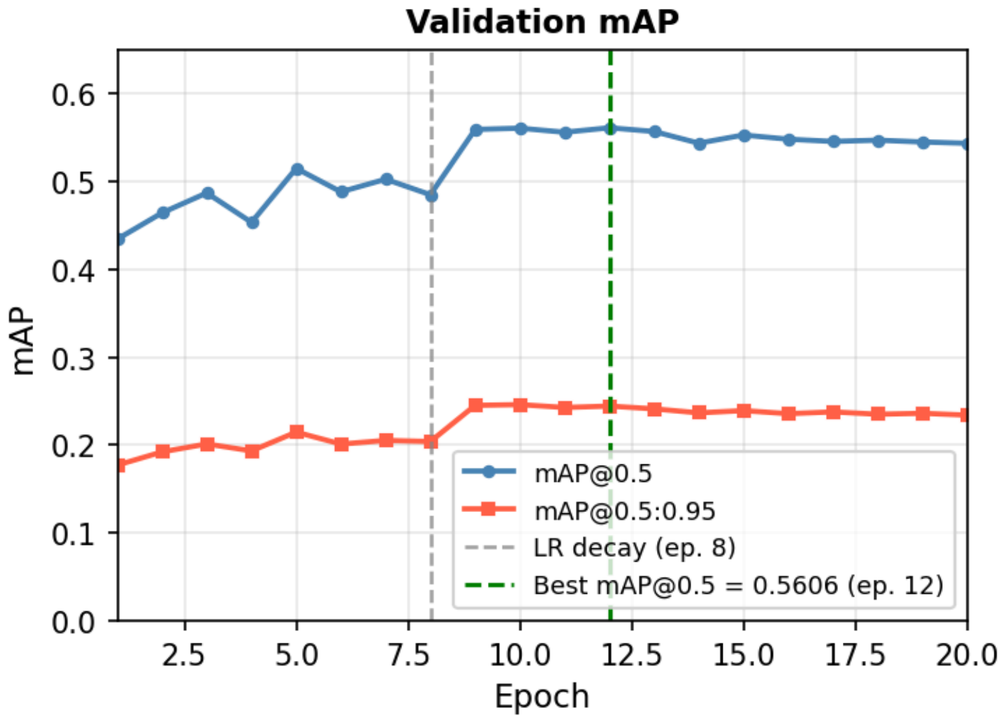
EXPERIMENT 2: FASTER R-CNN WITH RESNET-50 FPN.

- Model: Faster R-CNN + ResNet-50 FPN
- Improved best mAP@0.5 to 0.5606
- Better feature representation and localization than SSD
- Too slow for practical live deployment

EXPERIMENT 2

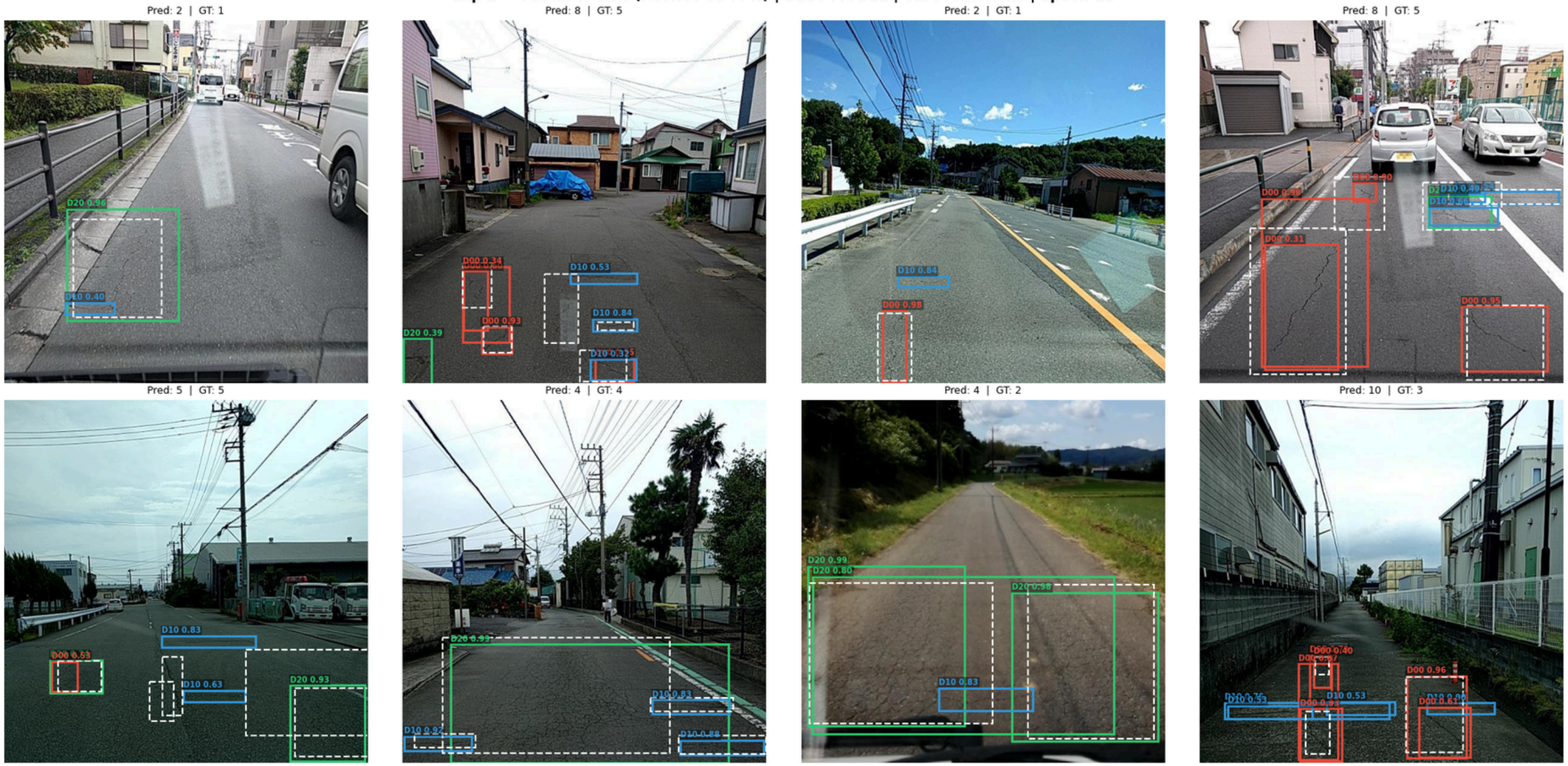
Faster R-CNN (ResNet-50 FPN) — Training Dynamics over 20 Epochs





EXPERIMENT 2

Exp 2 — Faster R-CNN (ResNet-50 FPN) | BEST MODEL | Val Predictions | Epoch 12



EXPERIMENTS

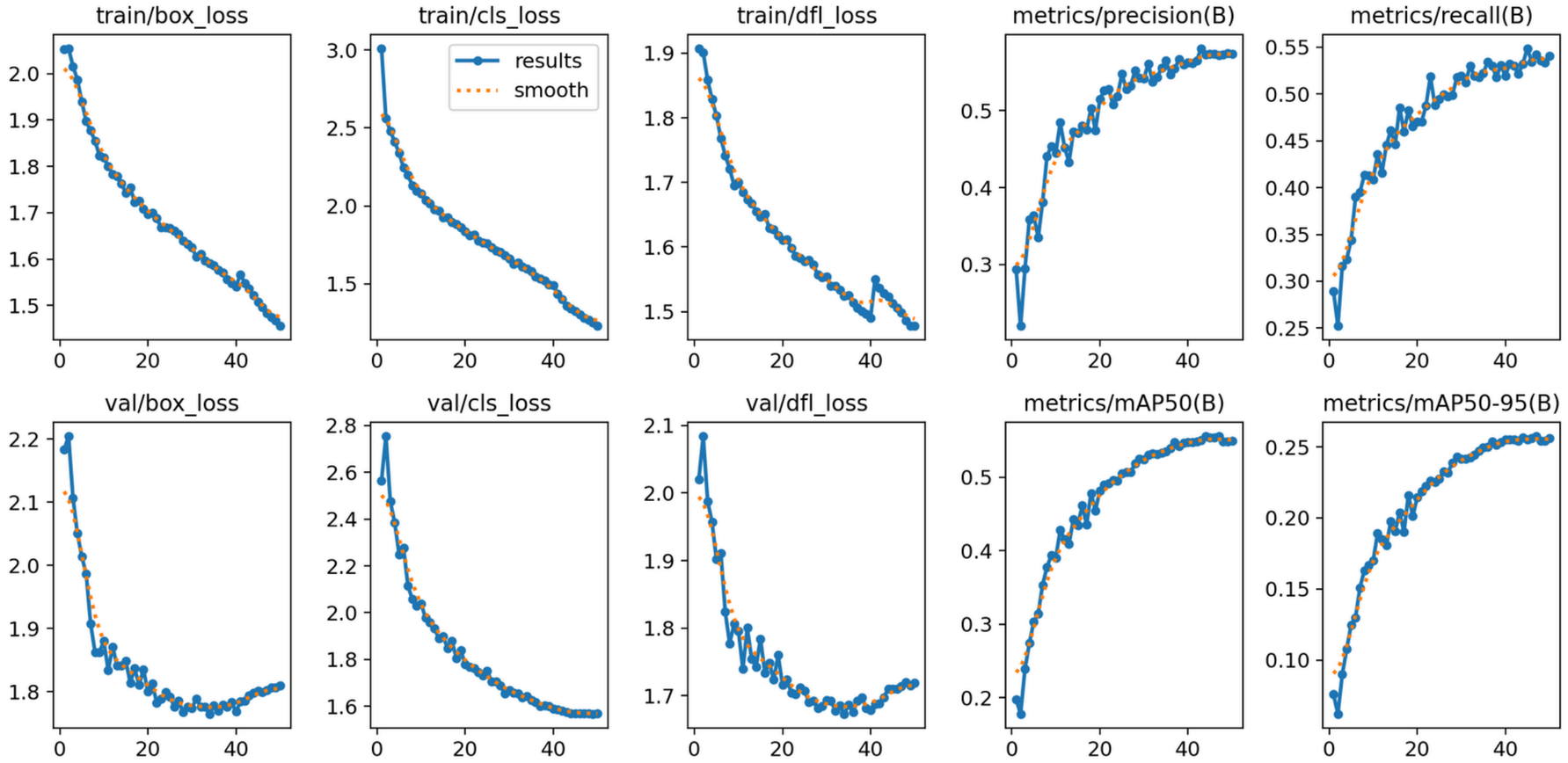
EXPERIMENT 3: YOLOV8S.

- Model: YOLOv8s
- Best mAP@0.5 = 0.5555
- Accuracy close to Faster R-CNN
- Much faster inference speed
- Provided per-class AP values for detailed evaluation

EXPERIMENT 4: ABLATION STUDY: INPUT RESOLUTION.

- Increased input size from 640×640 to 832×832
- Best mAP@0.5 = 0.5616
- D10 AP improved by +0.023
- Confirmed that higher resolution helps thin crack detection

EXPERIMENT 3



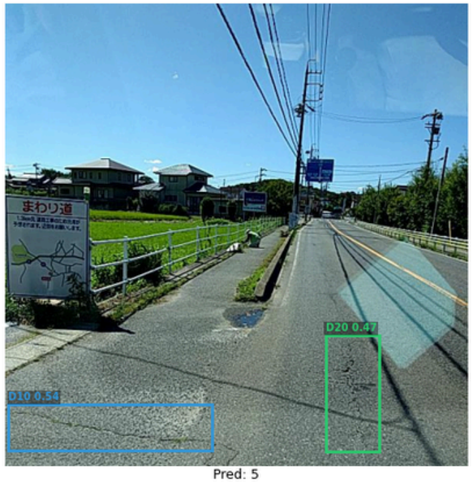
EXPERIMENT 3

Exp 3 — YOLOv8s | Val Predictions | Best Model
mAP@0.5: 0.5555 | mAP@0.5:0.95: 0.2578

Pred: 0



Pred: 2



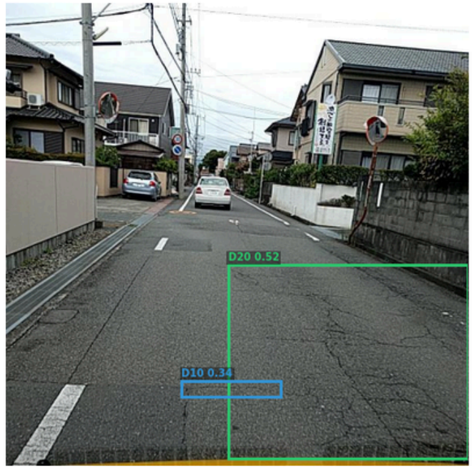
Pred: 3



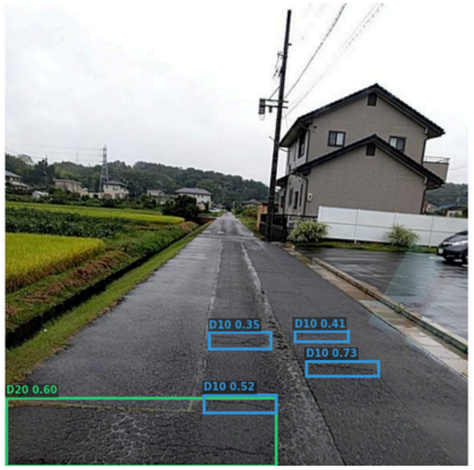
Pred: 1



Pred: 2



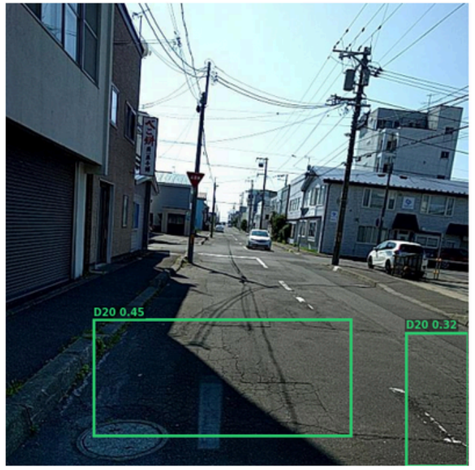
Pred: 5



Pred: 2



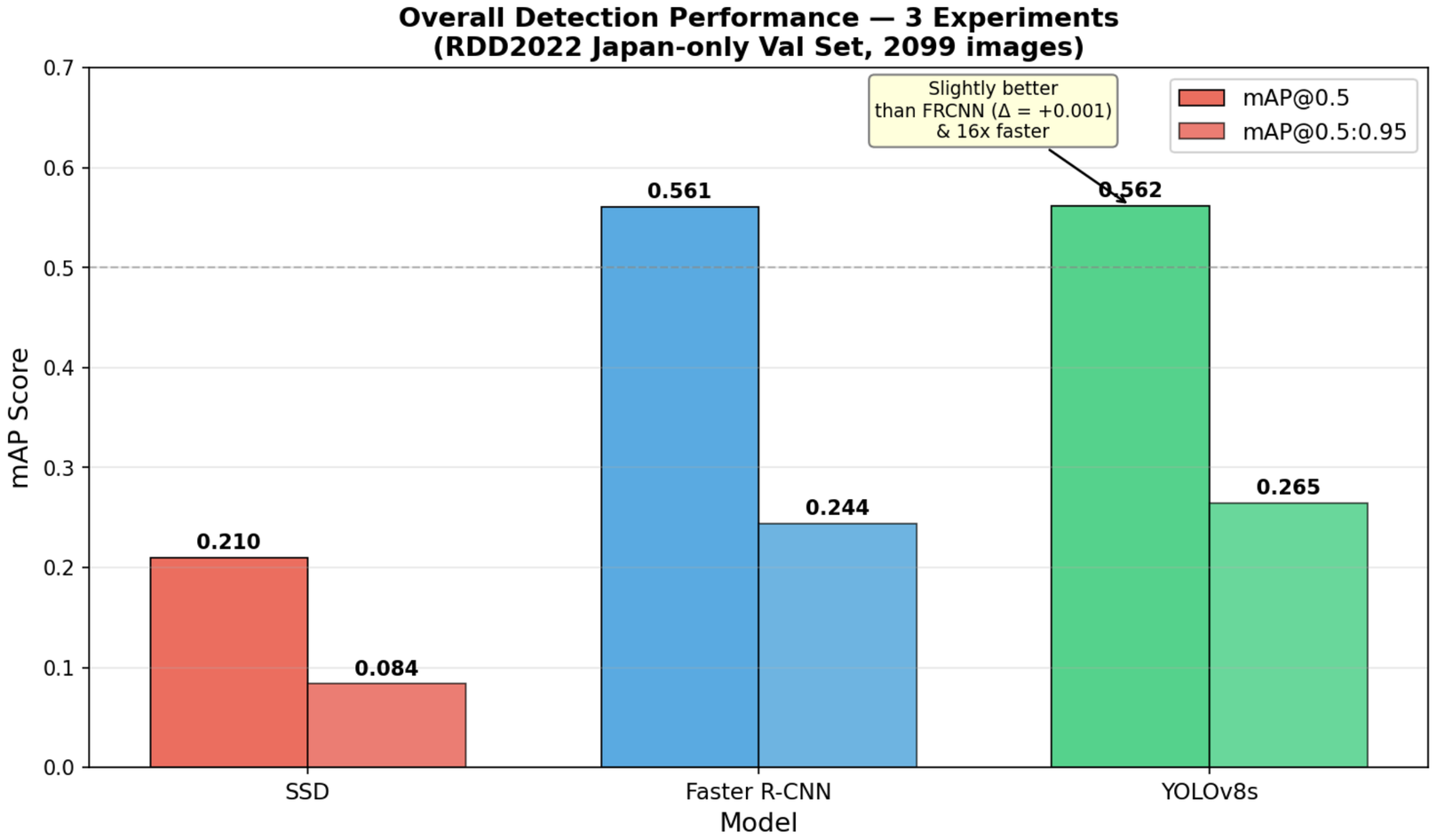
Pred: 2



RESULTS

- SSD baseline: mAP@0.5 = 0.210
- Faster R-CNN: mAP@0.5 = 0.561
- YOLOv8s Run A: mAP@0.5 = 0.556
- YOLOv8s Run B: mAP@0.5 = 0.562
- The Final Model: YOLOv8s Run B achieved the best mAP@0.5 = **0.562.**
- Key Achievements: The final inference speed reached 11 ms per image. Additionally, higher resolution improved D10 detection by +0.023 AP

RUSULT


Overall Detection Performance — 3 Experiments
(RDD2022 Japan-only Val Set, 2099 images)
mAP Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.210
0.084
0.561
0.244
0.562
0.265
Slightly better
than FRCNN (Δ = +0.001)
& 16x faster
mAP@0.5
mAP@0.5:0.95
SSD
Faster R-CNN
YOLOv8s
Model

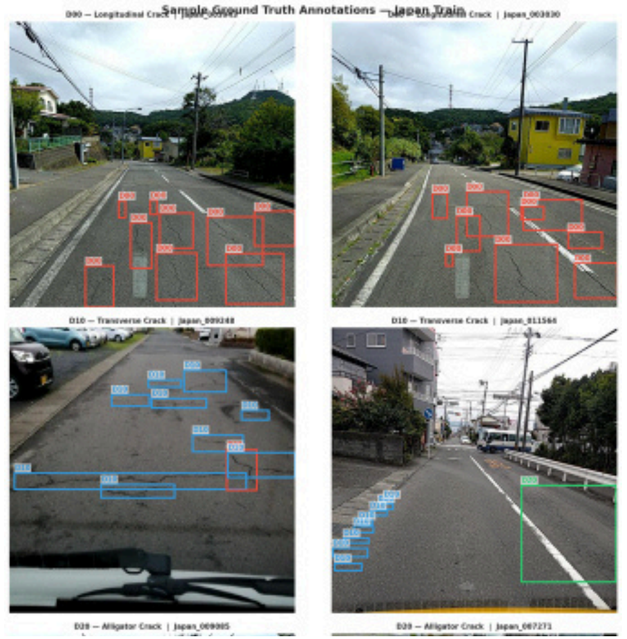
RUSULT

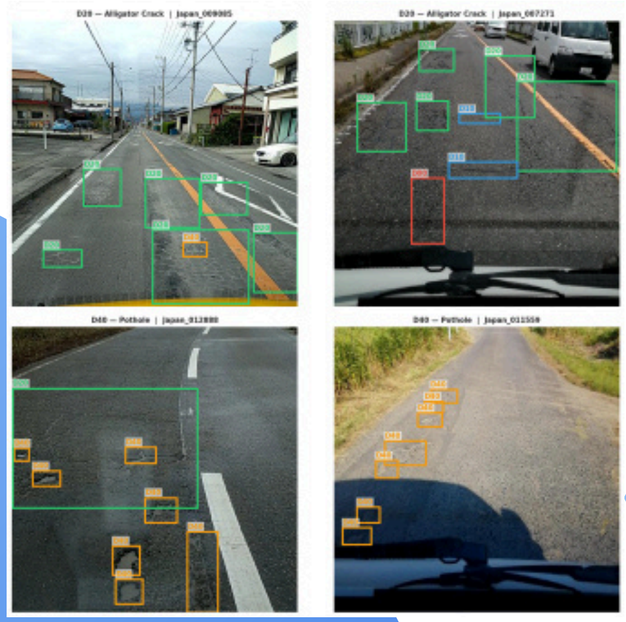
3-Experiment Summary — Road Damage Detection (RDD2022 Japan)

Metric	Exp 1: SSD	Exp 2: Faster R-CNN	Exp 3: YOLOv8s (Run B)
Architecture	SSDLite320	ResNet-50 FPN	YOLOv8s
Type	1-stage anchor	2-stage anchor	1-stage anchor-free
Params	5.1M	41.8M	11.1M
Input size	320px	~800px (FPN)	832px
mAP@0.5	0.2097	0.5606	0.5616
mAP@0.5:0.95	0.0836	0.2442	0.2648
Localization ratio	0.400	0.436	0.472
Speed (ms/img)	~10ms	~175ms	~11ms
D00 AP@0.5	N/A	N/A	0.471
D10 AP@0.5	N/A	N/A	0.461
D20 AP@0.5	N/A	N/A	0.701
D40 AP@0.5	N/A	N/A	0.612
Deploy ready?	No (weak accuracy)	No (slow 175ms)	Yes — Best choice

DEMO

- Deployed YOLOv8s Run B as a Gradio web application
- Supports image upload, video processing, and live webcam inference
- Uses OpenCV for bounding box annotation





CONCLUSIONS

This project shows that an iterative, evidence-driven approach is effective for improving road damage detection. YOLOv8s Run B provided the best balance of speed and accuracy, although thin crack detection and localization precision remain challenging.

Future work: could extend the system through multi-country training, better augmentation, and larger YOLOv8 models.

THANK YOU